
Network Security Assessment & Scanning Report

Project: ApexPlanet Internship Task 2

Target Host: 192.168.56.101 (Metasploitable2)

Assessor: Nikitha

ID:APSPL2631294

Date: April 23, 2026

1. Introduction

This report details the findings of a comprehensive network security scan performed on a target virtual environment. The goal was to identify open ports, determine the version of running services, and detect known vulnerabilities using the Nmap Scripting Engine (NSE).

Scope

- **Target IP:** 192.168.56.101
 - **Tools Used:** Nmap 7.94SVN, Nmap Scripting Engine (NSE)
 - **Methodology:** Passive reconnaissance followed by active service discovery and vulnerability scanning.
-

2. Executive Summary

The target machine, **192.168.56.101**, is classified as **EXTREMELY VULNERABLE**. The scan discovered **30 open TCP ports**, many of which host services with high-severity vulnerabilities, including pre-installed backdoors and unencrypted communication protocols. An attacker could achieve full administrative (root) access with minimal effort.

```
(kali㉿kali)-[~]
└─$ sudo nmap -sS -sV -O -p- -oN nmap_report.txt 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-23 01:26 EDT
Nmap scan report for 192.168.56.101
Host is up (0.00038s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
```

- **What to capture:** Your Kali terminal showing the command: `nmap -sS -sV -O 192.168.56.101`.
- **Caption:** Figure 1: Initial Service Discovery and OS Detection scan.

3. Detailed Technical Findings

A. Operating System Detection

- **Device Type:** General Purpose
- **Running:** Linux 2.6.X
- **OS Details:** Linux 2.6.9 - 2.6.33

B. Port and Service Table

Port	Protocol	Service	Version	Risk Level
21	TCP	FTP	vsftpd 2.3.4	CRITICAL
23	TCP	Telnet	Linux telnetd	HIGH

80	TCP	HTTP	Apache httpd 2.2.8	HIGH
1524	TCP	Bindshell	Metasploitable root shell	CRITICAL
3306	TCP	MySQL	5.0.51a-3ubuntu5	MEDIUM
6667	TCP	IRC	UnrealIRCd	CRITICAL

4. Critical Vulnerabilities & Exploitation Analysis

Vulnerability 1: vsFTPD 2.3.4 Backdoor (CVE-2011-2523)

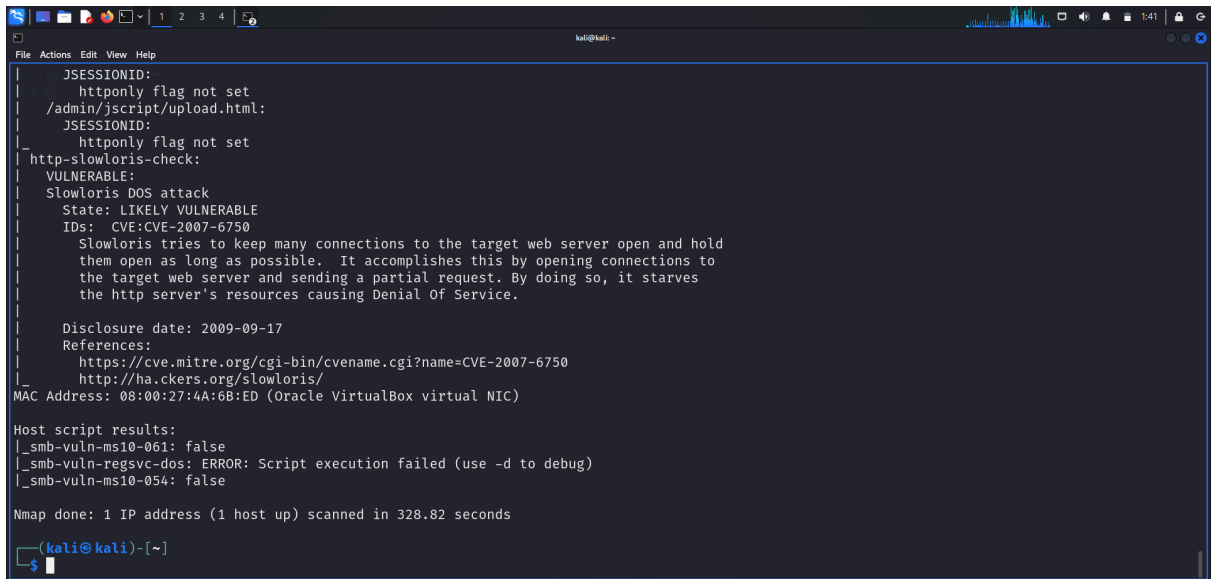
- **Description:** The FTP service on port 21 contains a malicious backdoor.
- **Impact:** Allows an attacker to trigger a root shell.
- **Evidence:** The NSE script `ftp-vsftpd-backdoor` confirmed the state as **VULNERABLE (Exploitable)**.

```

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 144.97 seconds

(kali@kali)-[~]
$ sudo nmap --script vuln 192.168.56.101 -oN nmap_vuln_results.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-23 01:35 EDT
Nmap scan report for 192.168.56.101
Host is up (0.00051s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|   vsFTPD version 2.3.4 backdoor
|   State: VULNERABLE (Exploitable)
|   IDs: CVE:CVE-2011-2523  BID:48539
|   vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|   Disclosure date: 2011-07-03
|   Exploit results:
|   Shell command: id
|   Results: uid=0(root) gid=0(root)
|   References:
|   http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|   https://www.securityfocus.com/bid/48539
|   https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|   https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp

```



```
File Actions Edit View Help
JSESSIONID:
  httponly flag not set
/admin/jscript/upload.html:
JSESSIONID:
  httponly flag not set
http-slowloris-check:
VULNERABLE:
  Slowloris DOS attack
  State: LIKELY VULNERABLE
  IDs: CVE:CVE-2007-6750
  Slowloris tries to keep many connections to the target web server open and hold
  them open as long as possible. It accomplishes this by opening connections to
  the target web server and sending a partial request. By doing so, it starves
  the http server's resources causing Denial Of Service.

  Disclosure date: 2009-09-17
  References:
    https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
    http://ha.ckers.org/slowloris/
MAC Address: 08:00:27:4A:6B:ED (Oracle VirtualBox virtual NIC)

Host script results:
_smb-vuln-ms10-061: false
_smb-vuln-regsvcs-dos: ERROR: Script execution failed (use -d to debug)
_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 328.82 seconds
(kali@kali)-[~]
$
```

- **What to capture:** The section of your `nmap_vuln_results.txt` that shows `ftp-vsftpd-backdoor: VULNERABLE`.
- **Caption:** Figure 2: Confirmation of the vsFTPD 2.3.4 Backdoor vulnerability.

Vulnerability 2: HTTP SQL Injection & Slowloris

- **Description:** The web server is vulnerable to Denial of Service (Slowloris) and multiple SQL Injection points on the Mutillidae application.
- **Impact:** Database theft and service downtime.

- **What to capture:** The terminal output showing the long list of "Possible sqli for queries" under Port 80.
- **Caption:** Figure 3: SQL Injection endpoints identified on the web application.

5. Recommendations

1. **Disable Insecure Services:** Immediately shut down Port 23 (Telnet) and Port 1524 (Bindshell). These protocols send data in plain text or provide direct shell access without authentication.
2. **Patch Management:** The versions of `vsftpd` and `UnrealIRCd` are trojaned. They must be updated to the latest secure versions or replaced with alternative software.
3. **Firewall Implementation:** Restrict access to ports like 139/445 (Samba) and 3306 (MySQL) so they are only accessible from trusted internal IP addresses.

4. **Web Application Hardening:** Sanitize input fields on the web applications to prevent SQL injection and update the Apache server to mitigate Slowloris attacks.

6. Conclusion

The scan results indicate that the target system is in a highly insecure state. The combination of backdoored services and unencrypted management protocols makes it a primary target for exploitation. Immediate remediation of the critical vulnerabilities (Ports 21, 1524, and 6667) is required.
